How Certain Is the Chatbot Arena Leaderboard? Multiplicity, Rank Spreads, and Sparse Category Effects

STAT 236 Final Project Proposal (Draft)

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Project type: Implementation and comparison of existing methods, and an application of multiple-testing / Higher Criticism methods to real data.

1 Background and Motivation

1.1 Chatbot Arena

Chatbot Arena (now LMArena, arena.ai) is the most widely cited leaderboard for large language models (LLMs). Users submit a prompt, two anonymous models respond, and the user votes for the better answer; model names are revealed only after the vote. As of Oct 8, 2026, the Text leaderboard ranks 414 models using about 8.7 million votes, with 29 category leaderboards (e.g., Coding, Math, Creative Writing, Hard Prompts, Language). How to rank chatbots from such pairwise data is problem (f) in Lecture 1 of this course.

1.2 The original methodology (Chiang et al., 2024)

Arena began as an academic project of the LMSYS group, and its statistical foundation is the team's own methodology paper (Chiang et al., 2024, ICML), which Arena's FAQ still cites as "our methodology." The paper:

- replaces the earlier Elo ratings with **Bradley–Terry (BT)** coefficients fit by maximum likelihood, because BT is better suited to statistical estimation (Sec. 4);
- computes confidence intervals by either the bootstrap or **sandwich** robust standard errors, and adopts the sandwich intervals after a simulation study (Sec. 5, App. A);
- defines an **approximate ranking** from an M-dimensional confidence set $\mathcal{C}$:

$$R_m = 1 + \#\{m' : \inf \mathcal{C}_{m'} > \sup \mathcal{C}_m\};$$

- states explicitly that this ranking **requires multiplicity correction**, because the ranking depends on all scores at once and is valid only if all of them lie in their intervals simultaneously. For this it uses a χ^2 **simultaneous confidence set** based on the full covariance matrix. Because the corrected intervals are wider, the paper reports both corrected and uncorrected intervals (Sec. 5, Sec. 7.1).

1.3 How the methodology evolved: blog posts and current code

Since the paper, Arena has become a company and changed its pipeline, documented in blog posts and an open-source package:

- **"Arena's Ranking Method" blog (Nov 2025, updated Feb 2026).** Introduces the **rank spread**. Previously the site reported only the "optimistic" rank, which is exactly the paper's R_m. The site now reports an interval $[R_m^{\text{best}}, R_m^{\text{worst}}]$:

$$R_m^{\text{best}} = 1 + \#\{m' : L_{m'} > U_m\}, \qquad R_m^{\text{worst}} = 1 + \#\{m' : U_{m'} > L_m\},$$

where $[L_m, U_m]$ is model m's confidence interval. Two models are "tied" if their rank spreads overlap. For example, the current #2 model has rank spread 2–10. The blog says the scores and confidence intervals themselves are unchanged, and **does not mention multiplicity correction or the χ^2 set**.

- **"Arena-Rank" blog and package (Dec 2025).** Arena open-sourced `arena-rank`, which it states "powers all of the leaderboards on our site today." The release adds reweighting for models with fewer battles, style control, and closed-form intervals in place of the bootstrap.
- **What the current code does.** In the latest version of `arena-rank` (GitHub `lmarena/arena-rank`, commit of Aug 3, 2026), every model class computes

$$\hat{\beta}_m \pm z_{1-\alpha/2}\,\widehat{\mathrm{SE}}_m, \qquad \alpha = 0.05,$$

where $\widehat{\mathrm{SE}}_m$ comes from the diagonal of the sandwich covariance matrix (`bradley_terry.py`, lines 186–195). These are **marginal, per-model 95% intervals**. The repository contains no multiplicity correction (no Bonferroni, χ^2, or simultaneous intervals), and the off-diagonal covariances are not used. The rank-spread computation itself is not in the repository.

1.4 The gap

The rank spread extends the paper's ranking formula, but the paper's validity argument assumed simultaneous (χ^2-corrected) intervals, while the current code produces marginal intervals, and no public document explains the change or its effect on the rankings. (We could not determine whether the live site ever displayed the corrected intervals between 2024 and now.) A rough check supports the reading that the site shows uncorrected intervals: for a model with 79,415 votes, the approximate marginal half-width is ± 2.4 points and the χ^2-corrected half-width is about ± 26, while the site displays ± 3.

Plugging marginal intervals into the rank-spread formula creates two errors in **opposite directions**, and their net effect on the leaderboard is unknown:

1. **No multiplicity correction makes intervals too narrow.** With $\sim$400 marginal 95% intervals, roughly 20 true scores are expected to fall outside their intervals. The rank spread depends on *all* intervals at once, so its joint guarantee may be far below 95%.
2. **The CI-overlap rule makes ties too frequent.** With equal standard errors and independent estimates, two non-overlapping 95% intervals correspond to a difference test at roughly the 99.4% level, which is stricter than needed. The rule also ignores correlation between estimates.

A second open question concerns the category leaderboards. Each is fit separately, and we found no test of whether a model's rank change between leaderboards (e.g., 7th overall but 3rd in Coding) is real or noise. With hundreds of (model, category) pairs, some apparent "specialties" will arise by chance.

2 Research Questions

The project asks two connected questions about **ranks within a leaderboard** (Q1 diagnoses, Q2 repairs), and one question about **differences between leaderboards** (Q3).

Q1 (Diagnosis): Is Arena's current rank spread reliable?

When Arena reports that a model's rank lies in, say, 2–10, how often is the true rank actually in that interval, for each model and for all models at once? Is the rank spread too wide, too narrow, or about right, given the two opposing errors above?

Q2 (Repair): Can multiplicity correction fix it, and at what cost?
Using the same evaluation as Q1, do corrected procedures reach the nominal 95% coverage? How much wider do the rank intervals become? Which procedure gives valid coverage with the narrowest intervals, and how does the real leaderboard change when we use it?

Q3 (Category effects): Are the "specialties" seen in category leaderboards real?
Answered in two steps, in this order:

1. **Is there any real specialty at all?** Do category leaderboards contain information beyond the overall leaderboard, or are their differences consistent with pure noise? (Global test: Higher Criticism, compared with min-P and χ^2.)
2. **If so, which ones?** Which specific (model, category) pairs have a real specialty? (Identification: Bonferroni and Benjamini–Hochberg.)

3 Methods

3.1 Data

Both datasets are publicly available on Hugging Face (organization `lmarena-ai`):

Dataset	Size	Fields used
`arena-human-preference-140k`	∼136K battles (2025)	`model_a`, `model_b`, `winner`, `category_tag`, `language`, `is_code`, `timestamp`, response length / formatting metadata
`leaderboard-dataset`	∼2.48M rows (updated Oct 2026)	Historical leaderboard snapshots: `model_name`, `score`, `score_ci_lower`, `score_ci_upper`, `observation_count`, `rank`, `category`, `leaderboard_publish_date`

We use the official `arena-rank` package so that our "Arena method" is exactly the code behind the public leaderboard.

Step 0 (verification). Before Q1, we confirm that we understand Arena's pipeline: (a) recompute rank spreads from the published intervals in `leaderboard-dataset` using the blog's formula and check that they match the site; (b) fit `arena-rank` on the 140K battles and check that the intervals are marginal (±1.96 SE), not corrected.

3.2 Q1 and Q2: one shared evaluation

Q1 and Q2 use the **same simulation**. In each simulated dataset we compute Arena's uncorrected rank spread (Q1) and the corrected alternatives (Q2) side by side, and compare all of them against the same known truth.

Why simulation. On real data nobody knows the true ranks, so the website's rank spreads cannot be checked directly. What can be checked is the *method*: in a world that resembles Arena and where the truth is known, how often does the method produce intervals that contain the true

ranks? This is like calibrating an instrument on a sample of known concentration before trusting its readings on patients.

Semi-synthetic design. Each battle has two parts: the matchup (who played whom) and the outcome (who won).

1. **Set the truth.** Fit BT on the real 140K battles. Treat the fitted scores β^* as the true strengths, and their ordering as the true ranks. This is done once.
2. **Keep the real matchups, regenerate outcomes.** Keep every real matchup (which pairs played and how often), discard the real outcomes, and draw new outcomes from the BT model with strengths β^*. This preserves Arena's real sparsity and imbalance (popular models with tens of thousands of battles, niche models with very few).
3. **Re-estimate.** On the simulated outcomes, refit BT and compute:
   - **Arena's method (Q1):** marginal 95% sandwich intervals via `arena-rank`, then rank spreads by the blog's formula;
   - **Corrected procedures (Q2):** see the table below.
4. **Compare with the truth.** For every procedure, record whether each model's rank interval contains its true rank, and the interval width.
5. **Repeat** steps 2–4 several hundred times.

Procedures compared in Q2.

Procedure	Idea	Multiplicity correction?
Bonferroni intervals	Replace $z_{1-\alpha/2} \approx 1.96$ by $z_{1-\alpha/(2M)}$ (≈ 3.9 for $M = 400$)	Yes; simple but conservative
χ^2 simultaneous set	$\{\beta : (\hat{\beta} - \beta)^\top \hat{V}^{-1} (\hat{\beta} - \beta) \leq \chi^2_{M-1,1-\alpha}\}$, the method of Chiang et al. (2024)	Yes; uses the full covariance, possibly very wide
Pairwise-difference tests	Test $H_0 : \beta_i = \beta_j$ for all pairs using $\mathrm{Var}(\hat{\beta}_i - \hat{\beta}_j)$ from the full covariance, control errors with Holm (FWER) or BH (FDR), and build rank intervals from the pairs declared different	Yes; also replaces the conservative CI-overlap rule with a direct test of each difference
Bootstrap rank intervals	Resample battles, refit, and take quantiles of each model's rank	No; a marginal benchmark for comparison

Evaluation metrics.

- **Marginal coverage:** fraction of models whose rank interval contains the true rank.
- **Simultaneous coverage:** fraction of simulations in which *all* rank intervals contain the true ranks.
- **Width:** average length of the rank intervals.

- **False orderings:** number of declared orderings "i ranks above j" that are wrong.

Robustness. Because β^* is itself an estimate, we also run fully synthetic variants (Erdős–Rényi comparison graphs; more or fewer models; denser or sparser graphs; larger or smaller gaps between models) to check whether conclusions hold across plausible "true worlds."

Application to real data (Q2). We apply the best-performing corrected procedure to the real data and compare the resulting rank intervals with Arena's published rank spreads: which models were wrongly separated, and which apparent ties can in fact be resolved.

3.3 Q3: category effects

Model. Let β_m be model m's overall strength and δ_{mc} its deviation in category c:

$$P(m \text{ beats } j \text{ in category } c) = \sigma\big((\beta_m + \delta_{mc}) - (\beta_j + \delta_{jc})\big), \qquad \sigma(x) = \frac{1}{1+e^{-x}}.$$

Specialties are relative, so we impose $\sum_m \delta_{mc} = 0$ within each category.

Hypotheses.

- *Individual*, for each (model, category) pair: $H_0^{(m,c)} : \delta_{mc} = 0$ (no specialty) vs. $H_1^{(m,c)} : \delta_{mc} \neq 0$ (stronger or weaker in category c).
- *Global*: H_0: all $\delta_{mc} = 0$ (category differences are pure noise) vs. H_1: some $\delta_{mc} \neq 0$ (possibly only a few).

Test statistic. For model m in category c,

$$z_{mc} = \frac{O_{mc} - E_{mc}}{\sqrt{V_{mc}}}, \qquad p_{mc} = 2\big(1 - \Phi(|z_{mc}|)\big),$$

where O_{mc} is the observed number of wins in category c, $E_{mc} = \sum \sigma(\hat{\beta}_m - \hat{\beta}_{\text{opp}})$ is the number expected from overall strengths, and $V_{mc} = \sum \sigma(1-\sigma)$ is its variance. With, e.g., 50 models and 10 categories, this gives $N = 500$ p-values.

Step 1: Is there any real specialty? (global test)

- **Higher Criticism** (Donoho & Jin, 2004). Sort the p-values $p_{(1)} \leq \cdots \leq p_{(N)}$ and compute

$$HC = \max_{k \leq N/2} \sqrt{N}\, \frac{k/N - p_{(k)}}{\sqrt{p_{(k)}(1-p_{(k)})}}.$$

  It compares, at every threshold, the observed number of small p-values with the number expected by chance, and is designed for the **rare and weak** regime, where only a few specialties exist and each is too small to be significant on its own.

- **Comparisons:** min-P (uses only the smallest p-value; best for few strong signals) and $\chi^2 = \sum z_{mc}^2$ (best for many weak signals).

Step 2: If so, which ones? (identification)

- **Bonferroni:** declare a specialty if $p_{mc} < 0.05/N$. Guarantees that the chance of *any* false specialty in the list is at most 5%; strict, so weak specialties may be missed.

- **Benjamini–Hochberg (BH):** with sorted p-values, find the largest k with $p_{(k)} \leq 0.10 \cdot k/N$ and declare the first k significant. Guarantees that on average at most 10% of the declared specialties are false; less strict, so it finds more.

Running both shows how strong the evidence is: Bonferroni gives the specialties we are nearly certain of, BH gives those that are likely. A possible outcome is that HC is significant while Bonferroni selects nothing, meaning specialties exist but are too weak to pin down individually.

Low rank plus sparse. We also arrange category scores into an $M \times C$ matrix, remove the low-rank "general ability" component by SVD, and apply the same two steps to the residuals.

Validation by simulation. As in Q1/Q2, we plant known specialties of varying number and size into semi-synthetic data and measure how often each method detects them. This also lets us interpret a null result on real data: it tells us how large a specialty could exist yet go undetected.

Dependence. The z_{mc} are correlated, because they share $\hat{\beta}$ and one model's deviation shifts its opponents' statistics. We calibrate null distributions by parametric bootstrap from the fitted BT model.

References

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